Predicting When Collection Statistics Help Top-k Selection: From Retrieval to Attention and Expert Routing

Team ANLP Doomsday

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IIIT Hyderabad, India https://github.com/avi1o1/ANLP-Doomsday

Abstract

Learned sparse retrieval, block-sparse attention and mixture-of-experts routing each select items by an inner product over learned non-negative features followed by a hard top-k. Only retrieval corrects this score, through the inverse document frequency (IDF), term saturation and length normalisation of BM25. We ask when the correction helps a learned representation and whether the answer can be predicted without relevance labels. We score fourteen sparse representations derived from one frozen encoder under all eight combinations of the three components on thirteen collections, in 182 primary and 1,226 sensitivity rows. The correction raises nDCG@10 in 90% of primary rows. An exact Shapley attribution assigns 61% of the gain to saturation, which reads no collection statistics. The pre-registered predictor, built on frequency diagnostics, does no better than always predicting a gain. Statistics of the index predict the saturation component. The value of IDF and length normalisation depends on relevance. In exploratory analysis only a pseudo-relevance proxy predicts it on held-out corpora (R^2 of 0.52 to 0.66). A decision rule built on that proxy raises mean nDCG@10 from 0.143 to 0.148, against 0.151 for an oracle. Adapters for attention and routing are implemented and will receive the frozen predictors next.

1 Introduction

A learned sparse retriever scores a query q against a document d as $\langle u_q, u_d \rangle$ over a vocabulary of non-negative features and returns the top k (Formal et al., 2021b; Nguyen et al., 2023). A block-sparse attention kernel scores KV page i for query q_t as $\langle q_t, c_i \rangle$ over page summaries and attends to the top-k pages (Tang et al., 2024). A mixture-of-experts router gates on $\langle h_t, w_e \rangle$ and sends the token to its top-k experts (Shazeer et al., 2017; Fedus et al., 2022). The three settings share one computation (Figure 1).

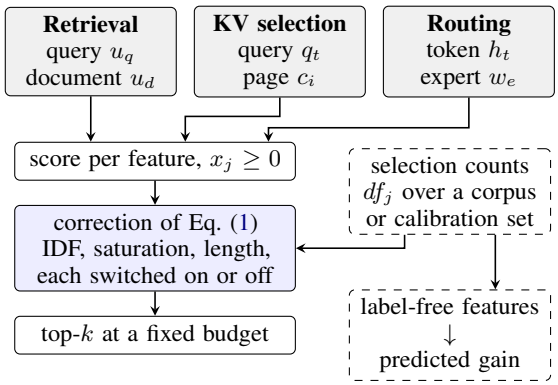


Figure 1: The computation shared by the three settings. Each scores an input against learned non-negative features and keeps the top k. The BM25 correction sits between the two steps. The selection counts it needs are also the input to the predictors of its gain.

Lexical retrieval learnt long ago to correct this score. BM25 down-weights features that occur across the collection, saturates repeated evidence and normalises for document length (Spärck Jones, 1972; Robertson and Zaragoza, 2009). Whether the correction helps learned features is unsettled. On one benchmark suite, BM25 over sparse-autoencoder features beats their inner product by 0.20 nDCG@10 while BM25 over SPLADE-v3 loses by 0.11 (Clavié et al., 2026). The published explanation is qualitative. BM25 helps while a vocabulary keeps a Zipfian frequency curve (Clavié et al., 2026; Han et al., 2026).

We make the question testable. From one frozen encoder we build fourteen sparse representations and score each under all eight combinations of IDF, saturation and length normalisation on thirteen collections, including three Indic ones. The retrieval setting is complete. The correction helps in 90% of primary rows. Saturation, the one component that reads no collection statistics, carries 61% of the gain. A predictor fixed in advance from frequency diagnostics fails. We trace the failure to the components themselves. The effect of saturation

is a property of the stored values, while the value of IDF and length normalisation depends on relevance. Section 2 states the problem formally and Section 7 reports the results.

2 Problem Statement

The correction. For query and document vectors x_q, x_d with non-negative coordinates we score

$$S(q,d) = \sum_j x_{qj}\, w_j\, T(x_{dj}, \ell_d). \qquad (1)$$

The weight w_j is one or the smoothed inverse document frequency $\mathrm{idf}_j = \ln(1 + (N - df_j + 0.5)/(df_j + 0.5))$, where df_j of the N documents have feature j. The length factor ℓ_d is one or $1 - b + bL_d/\overline{L}$ with $L_d = \sum_j x_{dj}$. The transform $T(f, \ell)$ is f/ℓ or the saturated $(k_1+1)f/(f+k_1\ell)$. Each component is a switch, so a configuration is a bit string $c \in \{0,1\}^3$ in the order IDF, saturation, length. Configuration `000` is the plain inner product and `111` is BM25 (Robertson and Zaragoza, 2009). IDF and length normalisation read collection statistics. Saturation reads only the document.

The unit of observation. Collection frequency belongs to a corpus, so one row of the analysis pairs a representation with a collection at a support budget (the query and document feature caps), BM25 parameters (k_1, b) and a basis seed. Each row carries eight nDCG@10 values Q_c, one per configuration.

What is predicted. The pre-registered target is the margin $m = (Q_{111} - Q_{000})/\operatorname{sd}(Q_{000}, \ldots, Q_{111})$, a standardised effect on the same scale in every setting. Section 7 shows that m is poorly shaped for prediction. We therefore add targets defined on the same eight values. The Shapley value ϕ_s of switch s is its marginal effect averaged over every order in which the switches can be turned on (Shapley, 1953). Writing Q_A for the configuration with the switches in A on,

$$\phi_s = \sum_{A \not\ni s} \frac{|A|!\,(2-|A|)!}{6} \big(Q_{A\cup\{s\}} - Q_A\big). \qquad (2)$$

The three values sum exactly to $Q_{111} - Q_{000}$. The collection gain $G = Q_{111} - Q_{010}$ is the value of IDF and length normalisation once saturation is applied. Its sign is the decision the project is about, whether to use collection statistics for a given representation and collection. A predictor maps label-free quantities to m, ϕ_s or G. It is scored by sign accuracy against the majority sign, by held-out R^2 and Spearman correlation. For G it is also scored by the nDCG@10 obtained when collection statistics are used only where the prediction is positive.

Transfer. Eq. (1) applies to any selection with a non-negative score and a selection count. For KV page i at decode step t, the feature value is the softplus of the Quest criticality $r_{t,i}$. Its frequency sf_i is the fraction of calibration queries for which the page enters the top-k among the pages that exist at that point. Its length is the norm of the page's concatenated key minima and maxima. For expert e at a token, the value is the router probability p_e, the frequency is the share of calibration tokens routed to e and the length is the norm of the router logits. The selected experts are combined with their original gate weights, so the correction changes only which experts run. In attention a row is a model, RULER task, context length and token budget, scored by RULER accuracy. In routing a row is a model and task, scored by negative token log-likelihood on WikiText and zero-shot MMLU accuracy. A predictor fitted in retrieval is applied there without refitting. The registered criterion is a sign accuracy of at least 70% and a mean absolute error of at most 1.5 times the corpus-held-out RMSE in retrieval. Always predicting a gain already reaches 90% sign accuracy in retrieval, so before any transfer run we add a third condition. The predictor must also beat the majority sign.

3 Motivation and Research Gap

The selectors above use the raw match score. They take no account of how often a feature is selected across the collection, how evidence accumulates within an item or how large the item is. Lexical retrieval corrects for all three with parts that arrived separately. These are inverse document frequency (Spärck Jones, 1972), term-frequency saturation (Robertson and Walker, 1994) and pivoted length normalisation (Singhal et al., 1996), combined in BM25 (Robertson and Zaragoza, 2009). Inside the transformer only fragments have been reinvented. Loss-free balancing adds a per-expert bias driven by recent load (Wang et al., 2024a), an additive frequency discount with no saturation or length term. Streaming inference keeps the first keys of the cache as an exception (Xiao et al., 2024).

Three papers have scored the same learned features with and without the correction, all in retrieval. BM25 wins on sparse-autoencoder features

of a dense retriever (Clavié et al., 2026) and on SAE concept units (Park et al., 2025). It comes out level on visual words (Han et al., 2026). Two of these papers give the Zipfian explanation. None of the three varies the vocabulary systematically or attributes the gain to the three components. The explanation has not been turned into a quantity that can be computed before the experiment, nor tested where the same computation runs inside a model.

4 Contributions

1. **A controlled grid.** Fourteen sparse representations from one frozen encoder are scored under all eight configurations on thirteen collections, with sensitivity families over support budget, k_1, b, basis seed and collection-statistic controls. The grid has 182 primary and 1,226 sensitivity rows. A second run on different hardware reproduces the primary rows it shares.
2. **An attribution of the gain.** Exact Shapley values show that saturation, a per-document transform, carries 61% of the gain. IDF alone carries about a tenth. Its share falls as the support budget grows.
3. **A negative result with its cause.** The preregistered predictor does no better than the majority sign. Frequency diagnostics describe the index, while whether IDF and length help depends on relevance.
4. **An exploratory label-free predictor.** Scoring each configuration against a reference retriever's ranking predicts the three components and the collection gain on held-out basis families and corpora.
5. **Transfer adapters.** KV-page selection and expert routing implement Eq. (1) and pass native-parity tests. They have not yet been run at scale.

5 Related Work

Term weighting. IDF was proposed as a statistical account of term specificity (Spärck Jones, 1972). Saturation came from approximating the two-Poisson model of within-document frequency (Robertson and Walker, 1994). Length normalisation was pivoted so that long documents are neither favoured nor penalised (Singhal et al., 1996). BM25 combines the three (Robertson and Zaragoza, 2009). Each part was motivated by word statistics, so it is open whether it helps features that are not words.

Retrieval scorers. Dense retrievers score by the inner product of embeddings (Karpukhin et al., 2020). Our encoder is multilingual E5 (Wang et al., 2024b). Learned sparse retrieval keeps the inverted index and learns term weights, through contextual exact match (Gao et al., 2021) or expansion over the subword vocabulary in SPLADE (Formal et al., 2021b,a; Lassance et al., 2024). Two frameworks place these methods side by side (Lin and Ma, 2021; Nguyen et al., 2023). Learned weights also change the cost of query processing (Mackenzie et al., 2021), which is why we report posting visits and index size. All of these methods score by inner product.

Learned concept vocabularies. Sparse autoencoders decompose dense activations into many non-negative features (Huben et al., 2024). Top-K SAEs fix the number active per input (Gao et al., 2025). In retrieval, CL-SR indexes SAE-decomposed embeddings as concept units weighted by IDF. Dropping that weighting costs 11 to 18 percent of its nDCG@10 on TREC-DL (Park et al., 2025), the one published ablation of the correction. SPLARE trains an SAE-based sparse retriever (Formal et al., 2026). SAE-SPLADE replaces the language-model head of SPLADE with an SAE (Zong et al., 2026). Both score by inner product. Quantised codes are the other route to a discrete vocabulary, through product quantisation (Jégou et al., 2011), residual quantisation (Chen et al., 2010) and their GPU implementations (Johnson et al., 2021).

BM25 on learned features. Clavié et al. (2026) show that SAE features of a frozen dense retriever have roughly Zipfian collection statistics and can be scored by BM25 directly. In their Appendix C, BM25 reaches 0.582 nDCG@10 against 0.381 for the inner product on SAE latent terms. On SPLADE-v3 it reaches 0.455 against 0.568. The scorers swap places when the vocabulary changes. Han et al. (2026) find the two within about two points of R@1 on image retrieval. They attribute the parity to top-k filtering already suppressing what IDF would down-weight. Raising their per-patch budget from 16 to 128 makes document frequencies converge and R@1 fall from 0.508 to 0.038.

Predicting effectiveness without labels. Query performance prediction estimates retrieval quality from the query and the ranking alone (Carmel and Yom-Tov, 2010). It uses the divergence of

the retrieved language model from the collection (Cronen-Townsend et al., 2002) or the spread of the top retrieval scores (Shtok et al., 2012). Retrieval systems can also be ranked without judgements by comparing each with pseudo-relevant documents drawn from pooled output (Soboroff et al., 2001). Our label-free predictor uses both ideas to predict a difference between scorers, a target neither line of work addresses.

Attributing an effect to components. The Shapley value assigns a joint outcome to players by averaging marginal contributions over orderings (Shapley, 1953). SHAP applies it to model predictions (Lundberg and Lee, 2017). Over a complete 2^3 factorial it is exact and needs no sampling.

Selection inside the transformer. Quest scores KV pages by an inner product against page min-max key summaries (Tang et al., 2024). H2O keeps recent and heavy-hitter tokens (Zhang et al., 2023b). SnapKV keeps the positions an observation window attends to (Li et al., 2024). StreamingLLM keeps the first keys as attention sinks (Xiao et al., 2024). Expected Attention prunes pairs that future queries are predicted to ignore (Devoto et al., 2025). Massive activations single out a few positions by magnitude (Sun et al., 2024). None of these uses the rate at which an entry is selected across queries. RULER measures long-context quality (Hsieh et al., 2024).

Expert routing. Sparse mixture-of-experts layers route each token to its top-k experts (Shazeer et al., 2017; Lepikhin et al., 2021; Fedus et al., 2022). Balance comes from an auxiliary loss (Zoph et al., 2022) or from a load-driven bias (Wang et al., 2024a). DeepSeekMoE splits experts more finely and adds shared experts (Dai et al., 2024). Su et al. (2026) find a handful of experts, identified by activation magnitude, whose removal degrades the model badly. We use OLMoE (Muennighoff et al., 2025), which activates 8 of 64 experts per layer. To our knowledge no work treats retrieval scoring, KV selection and routing as one scoring problem.

Viability. The correction costs one pass over the index. Sparse codes are cheap to build from one encoder. Label-free proxies for quality already work for single systems. The question therefore needs no new model to answer.

6 Method

Representations. All representations come from `multilingual-e5-base` (Wang et al., 2024b) at a pinned revision. Seven bases are fitted on the same 200,000 MS MARCO passages (Bajaj et al., 2016) and nowhere else. They are split-rectified identity, Gaussian projection (768), PCA (384), soft k-means over 4,096 centroids, product and residual quantisation (8×256) and a Top-K SAE with 8,192 features of which at most 32 are active. Pooled granularity applies each basis to the mean of the content token states. Token granularity applies it to each state before summing. This gives fourteen representations. A secondary SAE shares one dictionary across both granularities to separate aggregation from fitting. SPLADE has its own backbone, so it is an external comparison outside the grid.

Corpus statistics. Supports are capped before any statistic is computed, at 32 query and 128 document features in the primary budget. Document frequency, lengths and IDF come from the capped vectors of each target corpus. Retrieval is exact over the whole corpus. The primary grid uses $k_1 = 1.2$, $b = 0.75$ and seed zero. For a fixed linear map C and diagonal weights D, weighted scoring is bilinear, $(Cq)^\top D(Cd) = q^\top C^\top DCd$. It preserves every ranking of the inner product when $C^\top DC = \alpha I$ with $\alpha > 0$. Split rectification, truncation, saturation and length normalisation are not linear, so no representation here meets that condition. The code checks the identity and a counterexample numerically.

Collections. The thirteen collections come from eleven corpora. MS MARCO dev, TREC-DL 2019 and TREC-DL 2020 share one set of documents. The others are six BEIR sets (Thakur et al., 2021), LIMIT (Weller et al., 2026) and the Hindi, Bengali and Telugu portions of MIRACL (Zhang et al., 2023a), each evaluated within its language. LIMIT is built so that the top-k sets its queries need exceed what a low-dimensional embedding can represent.

Runs. The primary family (Table 1) ran on a four-GPU cluster node for all thirteen collections. A single-GPU workstation reran it for the ten collections outside MS MARCO with the cluster run's fitted bases and pinned data. The workstation also ran every sensitivity family.

Family	Rows
primary (14 representations × 13 collections)	182
support budget 16/64, 32/128, 64/256, native	462
$k_1 \in \{0.6, 1.2, 2.4\}$, $b \in \{0, 0.5, 0.75\}$	630
basis seeds 1 and 2	100
shuffled and uniform *df* controls	84
shared SAE dictionary	20
union on the ten non-MS MARCO collections	1,226

Table 1: Retrieval rows, each with eight scoring configurations. The primary count is the cluster run's. The other families and the union are the workstation run's. Families share their default cells, so the union is smaller than the sum. The MS MARCO budget rows are pending.

Significance. A paired bootstrap over queries recomputes numerator and denominator. Holm correction (Holm, 1979) runs within each pre-declared family, the all-on against all-off comparison and the three component main effects. The cluster run drew 2,000 samples, which puts the smallest attainable Holm p-value over 182 comparisons at 0.091. We therefore report significance from the workstation run, which drew 10,000.

Diagnostics. No feature describing a row reads relevance labels. The six pre-registered frequency diagnostics are the active fraction, the Zipf slope of the feature-frequency curve, the Gini coefficient of activation mass, frequency and activation mass in the most frequent 1% of features and IDF variance. The primary ridge model uses the slope, IDF variance and frequent activation share. Index statistics describe the stored values. The main one is cv_x, the coefficient of variation of the stored document values. Ranking features come from the stored top-100 list $R_c(q)$ of every configuration c and query q. They include how much each switch changes the top 10 and the shape of the top-100 scores (Shtok et al., 2012). The pseudo-quality $\tilde{Q}_c$ is the mean nDCG@10 of $R_c(q)$ when the top 10 of a reference retriever on the same query is taken as relevant. The references are text BM25, the dense encoder and the two fused by reciprocal rank. Replacing Q_c by $\tilde{Q}_c$ in Eq. (2) and in G gives pseudo-label estimates $\tilde{\phi}_s$ and $\tilde{G}$ with no fitted parameter.

Validation. Predictors are ridge regressions with the scaler and penalty chosen inside training folds. Folds hold out a whole basis family (linear, clustered, quantised, learned) or a whole corpus, reported separately. A fit on primary rows is also scored on the rows at other k_1 and b, which share collections with it.

Collection	BM25	E5	`000`	`111`	best `111`
MS MARCO dev	.222	.425	.013	.130	.318
TREC-DL 19	.456	.704	.036	.255	.604
TREC-DL 20	.484	.697	.035	.251	.556
ArguAna	.411	.445	.021	.130	.371
FiQA	.236	.382	.013	.085	.266
NFCorpus	.307	.325	.064	.115	.285
SCIDOCS	.152	.172	.009	.044	.120
SciFact	.662	.694	.089	.197	.419
TREC-COVID	.582	.695	.134	.246	.508
LIMIT	.939	.016	.001	.004	.032
MIRACL hi	.357	.583	.011	.124	.328
MIRACL bn	.407	.701	.025	.216	.490
MIRACL te	.331	.751	.019	.259	.579

Table 2: nDCG@10 of text BM25, the dense encoder, the mean over the fourteen sparse representations under the inner product (`000`) and full correction (`111`) and the best representation under `111`. The best is the token-level SAE on every collection except LIMIT, where it is token k-means.

Transfer adapters. The attention adapter implements Quest-style criticality over sixteen-token pages with dense prefill and sparse decode. It counts how often each page enters the top-k over causal prefill positions and applies Eq. (1) with pages for documents. The routing adapter applies the correction to OLMoE router probabilities for selection and combines with the original gate weights. In both, the all-off configuration calls the native code path. Six tests confirm exact native parity, causal calibration, budget charging including sink pages and the gate mixture.

7 Progress

7.1 Reproduction

The workstation run reused the cluster run's fitted bases and pinned data, so it reproduces the pipeline on second hardware. It is not an independent replication. Over the 140 primary rows the runs share, the median absolute difference in all-on nDCG@10 is 0.0007 (maximum 0.0077). The margin sign agrees in 136 of the 137 rows where it is defined. The differences come from fp16 rounding on different GPUs. Text BM25 agrees to seven decimals. Basis seeds move all-on nDCG@10 by a median standard deviation of 0.006 over 50 representation-collection groups.

7.2 Retrieval quality

The plain inner product of the capped sparse codes nearly fails (Table 2). Its mean nDCG@10 is be-

Rows	IDF	Sat.	Len.	IDF + len.
cluster, 182 primary	.10	.62	.28	.38 (.36–.41)
workstation, 140 primary	.10	.61	.29	.39 (.36–.42)
budget 16/64	.16	.60	.25	.40
budget 64/256	.08	.57	.35	.43
native, 3 collections	−.02	.55	.47	.45

Table 3: Share of the all-on minus all-off gain assigned to each switch by Eq. (2). Intervals resample whole collections. The primary budget is 32/128. The other budget rows cover the ten workstation collections. Native support covers ArguAna, NFCorpus and SciFact.

Basis	Gran.	111	cv_x	ϕ_{sat}	IDF + len.	$G > 0$
identity	pooled	.147	0.37	.043	.68	1.00
	token	.142	0.60	.060	.55	.77
Gaussian	pooled	.098	0.29	.020	.77	.85
	token	.088	0.55	.029	.64	.92
PCA	pooled	.222	0.75	.138	.24	.69
	token	.311	1.35	.155	.45	.46
k-means	pooled	.018	0.04	.000	.94	.31
	token	.185	1.71	.125	.11	.85
PQ	pooled	.012	0.00	.000	1.00	.62
	token	.144	1.44	.099	.09	.69
RQ	pooled	.056	0.00	.000	1.00	.92
	token	.292	2.10	.176	.18	.69
SAE	pooled	.127	0.18	.014	.66	.77
	token	.373	3.31	.193	.32	.46

Table 4: Per representation, mean all-on nDCG@10, mean saturation Shapley value, share of the gain from IDF and length and the fraction of collections where collection statistics help after saturation ($G > 0$), over the 13 primary collections. cv_x is the mean over the ten collections whose indexes were measured.

low 0.14 on every collection. Full correction raises the mean by a factor of 1.8 to 14. Token granularity is far stronger than pooled (mean all-on 0.219 against 0.097). The token-level SAE is the best representation on twelve collections and reaches 0.60 on TREC-DL 2019, against 0.70 for the dense encoder it comes from. On the ten workstation collections the pooled SAE reaches 0.095 with its own dictionary and 0.291 with the token-trained one, against 0.339 at token level. Most of the pooled deficit therefore comes from fitting the dictionary on pooled vectors. LIMIT behaves as its construction predicts. BM25 on words reaches 0.94, while the dense encoder and every representation built from it stay near zero.

7.3 Whether the correction helps

The margin is positive in 90% of the 179 primary rows where it is defined. Over the 140 workstation rows, Holm-corrected all-on against all-off comparisons are significant in 99. The main effects of saturation, length and IDF are significant in 93, 79 and 56 rows, with mean effects of +0.065, +0.032 and +0.013 nDCG@10. Full correction is the best of the eight configurations in only 43 of the 182 primary rows. The most frequent winner, in 51 rows, is saturation with length normalisation and no IDF. The all-off configuration scores below 0.01 nDCG@10 in 42% of workstation rows. The margin falls between 1.5 and 3.5 in 83% of them. It mostly records that the uncorrected scorer is broken, which leaves little variation to predict.

7.4 Where the gain comes from

Saturation carries about 61% of the gain and IDF about 10% in both runs (Table 3). Once saturation is applied, collection statistics add 0.015 nDCG@10 on average in the workstation run and 0.021 in the cluster run. They help in about two thirds of rows. As the budget grows from 16/64 to 64/256, the share of IDF falls from 16% to 8% while that of length rises from 25% to 35%. At native support IDF's share is below zero. Han et al. (2026) report the same trend for image features. On the five collections of the parameter family, all-on nDCG@10 at $b = 0$ falls as k_1 grows (0.170, 0.164, 0.153). At $b = 0.75$ it stays between 0.175 and 0.176. Weaker saturation therefore costs most when length normalisation is off.

Replacing document frequencies by shuffled ones on three collections turns the mean IDF Shapley value from +0.011 to −0.011. The real value is larger in 83% of 42 rows (one-sided Wilcoxon $p = 8.9 \times 10^{-7}$). Uniform frequencies remove IDF by construction and do not lower all-on nDCG@10 (0.157 against 0.147, $p = 0.15$). Real frequencies carry information, but little of it reaches nDCG@10.

Saturation's contribution rises with cv_x across representations (Table 4). The extreme case is exact. Pooled PQ and RQ store equal values with equal lengths ($\mathrm{cv}_x = 0$). Their Shapley values for saturation and length are exactly zero in all 26 of their cluster rows and all 166 workstation rows, as the mechanism in Section 7.6 requires. The share of the gain from collection statistics is largest for the weakest representations. For the two strongest, token-level SAE and PCA, collection statistics help after saturation on fewer than half the collections. That is where predicting the decision matters.

Target	Features	Family	Corpus	Params
ϕ_{sat}	pre-registered	$-.46$	.14	.02
	index	.47	.42	.50
	pseudo-labels	.56	.62	.76
ϕ_{len}	pre-registered	$-.30$	.14	.00
	index	$-.16$	.25	$-.02$
	pseudo-labels	.49	.70	.77
ϕ_{idf}	pre-registered	$-.01$	.00	.02
	index	.02	.02	.07
	pseudo-labels	.40	.21	.73
G	pre-registered	$-.07$	$-.02$	.16
	index	$-.05$	$-.09$	.15
	pseudo-labels	.38	.52	.62
	pseudo-labels, E5	.51	.66	.80

Table 5: Held-out R^2 of ridge regression fitted on 139 primary rows, holding out basis families or corpora, or predicting the 560 rows at other k_1 and b. Each tier adds features to the one above it. Pseudo-labels use text BM25 unless marked E5.

7.5 The pre-registered predictor

The primary ridge model predicts the margin with held-out R^2 of 0.04 across basis families and 0.02 across corpora in the cluster run. In the workstation run, with the rows that respond to fewer than two switches excluded, the values are -0.11 and 0.14. Its sign accuracy equals the majority-sign baseline in every case (0.899 and 0.924). The six-diagnostic model reaches $R^2 = 0.30$ across corpora in the workstation run but -0.54 across basis families. Its sign accuracy is never more than 0.03 above the majority sign. By the pre-registered criterion the frequency diagnostics carry no usable information about the margin.

7.6 What predicts the gain

Everything in this subsection was chosen after the pre-registered result and is exploratory.

Mechanism. With length normalisation off, saturation maps a stored value f to $g(f) = (k_1 + 1)f/(f + k_1)$. The ratio $g(f)/f$ falls as f grows, so saturation shrinks large values relative to small ones. If every stored value were equal, g would rescale all scores alike and change no ranking. The change grows with how unequal the values are, which cv_x measures. A smaller k_1 bends g more. How much saturation changes the ranking is thus a property of the index. Outside the two equal-valued codes the change helps in 91% of cluster primary rows. Length normalisation and IDF also change the ranking by an amount the index fixes. Whether the change helps depends on whether heavy documents are over-rewarded relative to their relevance and on whether rare features separate relevant documents from the rest. Both depend on relevance, which no statistic of the index measures.

Prediction. Index statistics predict saturation but fail for IDF and the collection gain (Table 5). The frequency diagnostics predict none of the targets. Pseudo-labels supply relevance information the index lacks and predict every target. The Shapley values and G are fixed linear combinations of the eight Q_c, so a pseudo-quality that orders the configurations like the true one yields correlated targets. The median within-row Spearman correlation between pseudo and true nDCG@10, over the eight configurations of 1,112 rows, is 0.98. With no fitting, the pseudo-label estimates rank the true values with Spearman 0.91 for saturation, 0.82 for length, 0.83 for IDF and 0.66 for G on the primary rows.

Decision. A rule with no fitted parameter uses collection statistics only where $\tilde{G}$ is positive. On the primary rows it raises mean nDCG@10 from 0.1429 (always using them) to 0.1478, against 0.1279 for never using them and 0.1510 for an oracle. The gain of 0.0049 is 61% of the 0.0081 available, with a collection-resampled 95% interval of 0.0003 to 0.0101. The rule recovers 67% at other budgets and 65% at other parameters. At other seeds it recovers 22% and the interval reaches zero. The gains are small in absolute terms, because always using collection statistics is already close to the oracle on most rows. The rule itself was chosen during the exploration.

Choice of reference. The main reference is text BM25. The correction under study is BM25 applied to other features. Agreement with BM25 may therefore reward configurations for behaving like BM25, whatever their relevance. Lexical systems also contributed to the judgement pools of several collections. The dense reference has the opposite weakness. The sparse codes come from the same encoder, so agreement with it partly measures how much of the embedding a configuration keeps. The prediction of G holds with the dense reference (R^2 of 0.51 and 0.66 in Table 5). With that reference the rule recovers 45 to 75% of the headroom across the four groups of rows. Neither reference is free of bias, so the pseudo-label result stays exploratory until it is tested on data fixed in advance.

Week	Dates	Work	Owner
1	3–9 Oct	Resume the MS MARCO budget sweep (84 rows) after the cluster maintenance. Run SPLADE-v3 as the external comparison and the conceptual replication of the Appendix C reversal. Validate both adapters at scale, with native parity, calibration cost and memory.	Anirudh
		Commit the second pre-registration, the pseudo-label rule on the unseen MS MARCO budget rows, the frozen transfer predictors and the transfer thresholds.	Mohit
2	10–16 Oct	Setting B on Llama-3.2-1B-Instruct over all 13 RULER tasks with 500 examples, 4K and 8K contexts, budgets of 512 and 1,024 tokens, with H2O.	Anirudh
		Second confirmatory round in retrieval.	Mohit
3	17–23 Oct	Setting C on OLMoE-1B-7B, WikiText-2 negative log-likelihood and zero-shot MMLU, with the loss-free-balancing analogue.	Anirudh
		Write-up of the bilinear identity and its ranking conditions. Draft of the final report from the retrieval results.	Arihant
4	24–28 Oct	Transfer scoring with uncertainty for both predictors, per setting, against the majority sign.	Mohit
		Figures and tables from the saved artifacts. Viva slides.	Arihant, Aviral
5	29–31 Oct	Final report	Arihant
		Slides and presentation. Code archive.	Aviral, Anirudh

Table 6: Schedule to the final submission. If time runs short we cut, in order, GSM8K and expert preservation, the 8K context in Setting B, MMLU in Setting C (keeping WikiText) and the Appendix C replication.

	Hypothesis	Result
H1	IDF + length share at default in $[.36, .42]$	.393
H2	ϕ_{sat} at $k_1 = 0.6$ exceeds $k_1 = 2.4$	$.093 > .080$
H3	within-corpus $\rho(\text{cv}_x, \phi_{\text{sat}}) \geq 0.3$	.89, .89
H4	cv_x model ranks ϕ_{sat} at $\rho \geq 0.5$	.87, .90
H5	$\phi_{\text{len}} = 0$ exactly at $b = 0$	max 0.0
H6	ϕ_{len} at $b = 0.75$ exceeds $b = 0.5$	$.045 > .029$

Table 7: Confirmatory tests. H2 and H6 have Holm-corrected $p = 3.6 \times 10^{-5}$. H3 and H4 are tested at $k_1 = 0.6$ and 2.4. H4's model is fitted on primary rows at $k_1 = 1.2$. H5 covers 210 rows.

7.7 Confirmatory tests

Six hypotheses were committed to the repository before their tests ran. All use the k_1 and b family, whose 630 rows the exploration had not used. H2 to H6 are Holm-corrected together. H1 uses the default cell, which coincides with primary rows, so it checks consistency only.

All six pass (Table 7). They are tests of the mechanism. Most are weak ones. H5 holds by construction, since the length factor is one at $b = 0$, so it checks the implementation. H2 and H6 test directions that follow from the form of Eq. (1). H3 and H4 carry the evidence. The value-dispersion predictor of saturation keeps its ranking at values of k_1 it was not fitted on. None of the six tests the pseudo-label predictor.

8 Remaining Plan

Table 6 gives the schedule. Anirudh Sankar writes and runs the experiment code. Mohit Kumar Singh analyses the results and owns the pre-registrations. Arihant Tripathy writes the report and relates the results to the literature. Aviral Gupta, who proposed the idea, makes the presentations. The pre-registered predictor is carried to attention and routing as registered. The pseudo-label predictor goes beside it, with a label-free reference native to each setting. For KV pages the reference is exact attention mass. For routing it is the dense softmax mixture over all experts. Both references come from the uncorrected model. No transfer run has been made yet. Both predictors and the thresholds will be frozen before any corrected outcome is seen. Each transfer result will be reported against the majority sign as well as the registered thresholds.

9 Limitations

All representations come from one encoder, so basis and encoder effects are not separable. E5 was trained with MIRACL data, so the Indic results do not concern unseen languages. The sparse codes are far below the dense encoder they come from, so the attribution may not hold for strong learned sparse retrievers such as SPLADE. The label-free predictor needs queries and a second retriever run. Its evidence is exploratory until the second pre-registered round. Routing is inference-time only, so nothing is claimed about training dynamics.

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